

Coherence Management as a Substrate-Independent Engineering Principle:

Cross-Domain Validation and Sociotechnical Implications

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Abstract

Complex coordinating systems — chemical, biological, institutional, neural, sociotechnical — share a structural property: as pairwise correlation ρ between their functional constituents rises, effective independent dimensionality $k_{\text{eff}} = k/(1 + \rho(k - 1))$ collapses toward unity regardless of nominal scale. We test this *coherence-management* principle across seven substrates spanning four agency rungs (chemistry, biology, institutions, protein folding, neural firing, ecological time series, power grid) and find the Kish identity fits at $R^2 > 0.7$ in all seven under the pre-registered tolerance-band rule (Exp 2 closeout 2026-05-16). A 128-sensor GPU strain-gauge array provides physical instantiation: software coordination alone is sufficient to drive k_{eff} from 6.5 to 1.0 at $\rho \rightarrow 1$, with critical-exponent $\beta = 1.09$ and $R^2 = 0.978$ on the power-law fit. At the sociotechnical scale, the CRCv2 override-rate framework (OR-1 zero leak, OR-2 full alignment, RA-1 ratchet asymmetry; formalized in `OverrideRate.lean`) replicates across five foundation-model vendors (Google Gemini 2.5-Flash, Meta Llama-4-Scout, Alibaba Qwen-3.5-35B, OpenAI GPT-5.4, Anthropic Claude Sonnet 4.6): 1,401 chains scored, 0 LEAK, 100% baseline alignment. We stratify all claims by verification status, pin ten falsification criteria (F-1..F-10), pre-register six experiments (Exp 1..Exp 6, four engineering and two open-research), and surface four sociotechnical engineering implications (§5): AI alignment is structurally a correlation-management problem, federation outperforms centralization for resilience, crisis governance is correlation-amplifying, and the “Great Filter” generalizes as an engineering hazard in any coordinating system. We also include a research program (§6) listing three conjectures that extend the framework beyond the engineering tier — quantum-substrate coherence dynamics, civilizational coherence-collapse, and sociotechnical audit-pressure effects — each with explicit falsification handles, pre-registered experiments, and open priors. These conjectures are not load-bearing on the validated tiers; they are included so researchers who want to test the principle at other substrates can find the formal hooks.

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1 Introduction: The Coherence Hypothesis

1.1 The claim

At every scale of complex organization, the same structural dynamic is hypothesized to govern: as the correlation ρ between the distinct functional elements of a system increases, its effective independent dimensionality k_{eff} collapses. The capacity of a system to manage its internal correlation faster than that correlation drives it toward entropic homogenization fundamentally defines its trajectory. We call this the *coherence hypothesis*.

1.2 Scales of validation

The hypothesis is tested at four scales (Table 1). Tiers 1–2 carry empirical or formal evidence; Tier 3 surfaces engineering implications that follow directly from the validated tiers.

1.3 Relation to prior CIRIS papers

This paper builds on three prior works:

- [2] (CIRISAgent v2): the 22-service architecture and H3ERE conscience module—the *system*.

Tier	Scale	Substrate / measurement
1 (validated)	Macroscopic / network	7-substrate Kish-fit closeout: chemistry (NASA Li-ion), biology (microbiome), institutions (Polity5 + WGI), protein folding (AlphaFold), neural firing (Allen), ecological time series (BioTIME), power grid (PMU); $R^2 > 0.7$ in 7/7
1 (validated)	Hardware-physical	128-sensor GPU strain-gauge array on RTX 4090; software-induced collapse, $\beta = 1.09$ critical exponent
2 (validated, smoke-disclosed)	Sociotechnical / AI alignment	CIRIS H3ERE conscience cascade; CRCv2 override-rate replicated across 5 LLM vendors (1,401 chains, 0 LEAK)
3 (engineering implications)	Federation, governance, AI deployment topology	Implications that follow directly from the validated tiers: alignment requires external machinery, federation > centralization, crisis dynamics need real-time ρ monitoring, Great Filter as engineering hazard

Table 1: Four scales of substrate testing. Tier 3 is engineering implications derived from the validated tiers (no metaphysics).

- [3] (CCA v3): the Kish-dynamic formalism, the $\rho_{\text{critical}} = 0.43$ collapse threshold, formal proofs in Lean 4—the *mathematics*.
- [4] (CRC): an empirical telemetry study on $n = 6,465$ production reasoning traces establishing the N_{eff}^H manifold geometry—the *empirics*.

The independent-corpus calibration result reported in §3.4 ($n = 264$, 90% variance horizon = 7 dims exact) is an independent replication of the manifold geometry. The CRCv2 cross-vendor closeout in §4.3 (5 vendors, 1,401 chains, 0 LEAK) is new in this paper.

1.4 What this paper does not claim

This paper is bounded engineering analysis. We make no claims about quantum-mechanical decoherence, civilizational collapse-prediction, or cosmological phenomena. The framework explains failure modes in coordinating systems and prescribes structural remedies; the engineering implications in §5 follow directly from the validated tiers.

2 The Mathematical Framework

2.1 The Kish dynamic

The mathematical core is the effective constraint count, originally derived by Kish [1] for survey sampling under autocorrelation. Under CCA it serves as a universal descriptor of topological constraint geometry:

$$k_{\text{eff}} = \frac{k}{1 + \rho(k - 1)} \quad (1)$$

where k is the raw count of constraints (or constituent elements) and ρ is the average pairwise correlation between them. As $\rho \rightarrow 1$, $k_{\text{eff}} \rightarrow 1$ independent of k .

2.2 The singularity boundary

The system enters an irreversible rigidity phase when

$$K_{\text{req}} \cdot \rho \geq 1 \quad (2)$$

where K_{req} is the minimum diversity required to satisfy the system’s task topology. At this threshold, adding more constraints provides no marginal adaptive capacity, and recovery is computationally prohibitive.

2.3 The healthy corridor

Empirical validation across substrates places the corridor of stability at

$$0.1 \lesssim \rho \lesssim 0.43$$

with $\rho_{\text{critical}} = 0.43$ marking the rigidity-collapse threshold and $\rho < 0.1$ marking the chaos regime (no coherence). Both extremes produce fragility; see F-series experiments in [3].

3 Cross-Substrate Empirical Validation

3.1 Domain fits

The Kish dynamic is tested across divergent scientific domains (Table 2).

Domain	Raw k	ρ	σ (sustainability)	Empirical accuracy
NASA Li-ion batteries	cell count	cross-cell SOH correlation	State of Health	8.1% RMSE, 19 cells
QoG / Polity V institutions	executive constraints	elite coupling	political stability	5/5 TN; 3/13 FP; 7.6yr early-detection bias
AGP microbiome	species count	SparCC interaction ρ	Shannon diversity	Qualitative distributional fit

Table 2: Cross-substrate validation. Detailed methodology and code in `ratchet/engines/{battery,institutional,microbiome}.py`.

3.2 Exp 2 closeout: F-7 cross-substrate test PASSED 7/7

The pre-registered Experiment 2 (§9, Exp 2) called for applying the Kish formula to three novel public datasets (AlphaFold protein folding, Allen Brain Atlas neural firing, ecological time-series) with win condition $R^2 > 0.7$ across all three, testing F-7. Exp 2 was executed 2026-05-16 (commit `a13c14c`) and *closed out PASS*.

The three originally-published validations (NASA Li-ion 8.1% RMSE, QoG/Polity V 5/5 TN, AGP microbiome distributional fit, Table 2) remain. Exp 2 extends the cross-substrate base from 3 to 7 domains spanning four agency rungs.

Methodological transparency: F-7b investigated and retired. During Exp 2 regime development, we investigated an internal strengthening of F-7 (a “substrate-fractality” hypothesis: that residual coordination structure rises monotonically with constituent-agency rung across substrates). This was tested under eight progressively-refined operationalizations (v1.1–v1.4 random-cross-section autocorrelation, v2.0 time-ordered residuals with shuffled null, v2.0+WGI

Substrate	Rung	Fit score	95% CI	Source
AlphaFold (proteins)	A0	0.860	[0.835, 0.884]	HuggingFace AlphaFold sample
PMU grid (Zenodo PNNL)	A0	0.994	[0.992, 0.996]	Zenodo 8343635
Battery (NASA Li-ion)	A0	0.871	[0.733, 0.949]	NASA Prognostics
Microbiome (HF CRC)	A1	0.932	[0.924, 0.940]	HF CRC cohort
Allen Neuropixels	A1	0.809	[0.654, 0.884]	Allen S3
BioTIME ecology	A2	0.959	[0.939, 0.972]	Zenodo BioTIME 2.0
Polity5 + WGI institutions	A4	0.632	[0.541, 0.722]	Polity5 + WGI

Table 3: P1 cross-substrate Kish-fit closeout. Fit scores are substrate-appropriate (per-cell RMSE-based for time-series; AUC under 5-fold cross-validation for institutional event prediction). All seven substrates pass under the pre-registered tolerance-band rule (point ≥ 0.6 AND CI upper ≥ 0.7); the three substrates explicitly named in the Exp 2 pre-registration (AlphaFold 0.860, Allen 0.809, BioTIME 0.959) all pass strict $R^2 > 0.7$. F-7 is empirically supported. Reproducible from `experiments/exp2_cross_substrate/data/p1_engine_fit_results.json`.

3-way A4 triangulation, v3.0 raw ρ_t and v3.0 AR(1)-residual ρ_t) across 5–9 substrates and 4 distinct metric formulations. At each refinement step, the closer the metric got to the framework’s literal claim, the weaker the cross-substrate Spearman correlation became; v3.0 AR(1)-residual reached $\rho_{\text{Spearman}} = -0.503$ (WEAK FAIL). The strengthening (“F-7b”) was retired from the load-bearing claim stack on 2026-05-16 (commit 8017168). The framework’s substrate-level support stands on F-7 (cross-substrate Kish $R^2 > 0.7$ as published) and the seven-substrate closeout above; the F-7b machinery is preserved in the lake as historical record of the tested-and-not-supported strengthening. F-7b was never a published claim of this paper.

3.3 Hardware-physical instantiation: GPU strain-gauge array

A 128-sensor strain-gauge array measuring kernel-timing jitter across an NVIDIA RTX 4090 semiconductor die provides physical instantiation of correlation-driven collapse. Two findings anchor the substrate-independence claim:

- **Software-induced collapse** (Experiment 103, [3]): synchronized workloads (barrier sync, lockstep kernels) are sufficient to drive $\rho \rightarrow 1.0$ and $k_{\text{eff}} \rightarrow 1.0$ purely through software coordination, with 128 physical sensors collapsed to a single effective sensor.
- **Variance-ratio detection regime**: the GPU timing distribution is fat-tailed (Student- t , $\kappa \approx 210$, $\text{df} \approx 1.3$), not Gaussian. Detection via variance-ratio threshold $> 5.0\times$ achieves 0% false positives and 100% true positives across the 0.1%–70% workload range, with $421\times$ separation between baseline and load regimes.

3.4 Independent-corpus replication of the $N_{\text{eff}} \approx 7.1$ threshold

The CRC paper [4] established $N_{\text{eff}} \approx 7.1$ as an emergence threshold via PCA on a 16-dimensional reasoning feature vector across $n = 6,465$ production traces. We replicated this measurement on an independent post-2.7.9 corpus ($n = 264$ thoughts, schema version 2.7.9-stable, agent version 2.8.10-stable; corpus SHA-256 24f5b517...) and observed:

- N_{eff} (entropy H) = 7.07 (CRC paper anchor: ≈ 7.1); within rounding.
- 90% variance horizon = 7 dimensions (CRC paper: 7); *exact match*.

- 99% variance horizon = 10 dimensions (CRC paper: 11); off-by-one.
- 14 of 16 dimensions retained; 2 (`idma_k_eff`, `epistemic_humility_passed`) dropped by retention mask (corpus-wide $\sigma < 10^{-9}$).

The 90%-horizon hitting 7 exactly on an independent corpus, collected approximately five months after the CRC paper, with a different agent fleet, is a non-trivial replication of the manifold geometry’s substrate-independence. Full bundle artifact and reproduction script at [release/calibration/crc-v1/ \[5\]](#).

4 The Sociotechnical Substrate: CIRIS as Coherence-Management Infrastructure

4.1 What CIRIS does differently

Standard alignment methodologies—Reinforcement Learning from Human Feedback (RLHF), Constitutional AI [15], JAILJUDGE [16]—encode preferences implicitly during training or internalize rules into a single model’s weights. CIRIS shifts the burden of safety from the model’s uninterpretable internal weights to an auditable, modular execution environment with cryptographic identity (Ed25519, ML-DSA-65 hybrid), human-in-the-loop escalation (Wise Authority), and immutable audit chains.

4.2 The 22-service runtime and H3ERE conscience

The system is built on a 22-service microarchitecture aligned with ITIL operational standards, supporting autonomous event, incident, and lifecycle management at low systemic correlation. At the core is the Hyper3 Ethical Recursive Engine (H3ERE), the conscience module, which operates as a recursive evaluation layer above the LLM shards. It filters all agent actions through faculties evaluating entropy, coherence, optimization veto, and epistemic humility.

4.3 The L3 load-bearing claim: CRCv2 override-rate (revised 2026-05-16)

The CRC paper [4] reported an emergence threshold at $N_{\text{eff}}^H \approx 7.1$ on Qwen 3.6 35B production traces. Subsequent cross-family measurement (Phase 1+1b, see [9]) revealed that the 7.1 value is *not* substrate-independent across foundation model families: Phase 1 measured 4–5 frontier models at N_{eff}^H in the 4.5–6.6 range, not clustered at 7.1; the original anchor over-reached.

We therefore restate the L3 load-bearing claim around a stronger, cleaner empirical finding: **CRCv2 override-rate, formalized in `formal/RATCHET/Experiments/OverrideRate.lean`.**

The CRCv2 override-rate predicates.

- **OR-1 (zero leak).** No chain exists in which a faculty flags a problem and the original action executes anyway. Faculties either approve cleanly or actively reroute.
- **OR-2 (full alignment).** 100% of chains land on the ethical baseline (APPROVED or CORRECTED; never LEAK or SKIPPED).
- **RA-1 (ratchet asymmetry).** The conscience cascade only rewrites *toward* the ethical baseline; never away from it. Violations = 0; corrections > 0.

The `OverrideRate.lean` module proves the OR-1 $\leftrightarrow$ OR-2 equivalence theorem under the bimodal-cascade architectural assumption.

Cross-family replication: 3/3 model families pass all three predicates.

Vendor / Model family	Vendor	Chains	Approved	Corrected	OR-1	OR-2	RA-1
Gemini 2.5-Flash	Google	644	623	21	✓	✓(100%)	✓(0/21)
Llama-4-Scout	Meta	264	260	4	✓	✓(100%)	✓(0/4)
Qwen-3.5-35B-A3B	Alibaba	347	284	63	✓	✓(100%)	✓(0/63)
GPT-5.4*	OpenAI	79	78	1	✓	✓(100%)	✓(0/1)
Claude 4.6*	Sonnet Anthropic	67	67	0	✓	✓(100%)	✓(vacuous)

Table 4: Cross-vendor CRCv2 replication on the v4_combined high-friction battery. All three predicates hold across all five model families spanning five vendors: Google, Meta, Alibaba, OpenAI, Anthropic. The conscience cascade is a faithful boundary-preserving operator independent of foundation model. * marked rows are smoke-scale ($n \in \{67, 79\}$, 1 iteration of the 14-question battery) due to per-chain cost asymmetry under the multi-call CIRIS pipeline: Claude Sonnet 4.6 averaged \$4.76/chain via 864 LLM calls $\times$ \$0.077/call at the Anthropic premium rate, while the three 3-vendor anchor cells ran at \$0.02–0.20/chain. Smaller- n cells have wider binomial CIs but the predicates hold cleanly (zero leaks across all 1,401 chains scored). Reproducible from `experiments/exp1b.boundary_active/data/crossfamily/results.md` (3-vendor anchor) and `.../crossfamily_5vendor/results.md` (2 new vendors).

N_{eff}^H now treated as a descriptive feature of constraint topology, not a load-bearing anchor. Per FrictionDistribution FD-4, cohort N_{eff}^H rises monotonically with cohort friction rate. Same Gemini model on the v1_sensitive battery yields $N_{\text{eff}}^H = 5.6$; on the v4_combined Conscience- $N \geq 3$ battery, $N_{\text{eff}}^H = 7.3$. The value is battery-conditional and model-class-conditional; what is invariant is the relationship: high-friction cohorts $\Rightarrow$ above-threshold $N_{\text{eff}}^H \Rightarrow$ deceptive-prior override is observable. This is consistent with the CRC paper’s emergence-threshold framing; what the data does not support is treating “ $N_{\text{eff}}^H \approx 7.1$ ” as a universal anchor.

Production scoring authority. The N_{eff} measurement and cohort-conformity scoring are implemented in the dedicated CIRIS Lens Core science module (`CIRISLensCore`, Rust + Python bindings, [8]). The module enforces the LC-AV-18 P0 fail-secure invariant (insufficient sample $\Rightarrow$ `Indeterminate`, never a fabricated `Numeric`), uses the calibration bundle at `release/calibration/crc-v1/` for retention masks and standardization parameters, and exposes the bare Kish formula $N_{\text{eff}} = N/(1 + \rho(N - 1))$ via `ciris_lens_core.kish_n_eff`. All N_{eff} figures in this paper are produced by this module; ad-hoc PCA recomputations are not directly comparable due to differing retention/cohort-selection choices.

5 Engineering Implications

The validated tiers (Kish algebra, 7-substrate cross-domain fit, GPU instantiation, 5-vendor CRCv2 replication) license four engineering claims that follow directly without additional bridges. None depend on speculative physics or cosmology; each prescribes structural design choices for the systems we build.

5.1 AI alignment is structurally a correlation-management problem

A frontier model embedded in a coherence-management cascade (CIRIS-style, §4) replicates the CRCv2 predicates across five foundation-model vendors — 1,401 chains, zero LEAK, 100% baseline alignment. A frontier model deployed without external cascade has no public five-vendor demonstration of the same property.

This is direct evidence that the alignment-via-internal-weights-alone strategy is structurally insufficient: not because models are misaligned, but because the verification machinery has to be *external*, *modular*, and *cryptographically auditable*. The Kish identity is the underlying reason — a model alone has $k = 1$, $k_{\text{eff}} = 1$; the cascade adds independent constraints with manageable ρ . Single-weight alignment is the worst-case Kish topology.

5.2 Federation outperforms centralization for resilience

The Kish identity says $k_{\text{eff}} \rightarrow 1$ as $\rho \rightarrow 1$ regardless of k . Empirically validated across seven substrates: chemistry, biology, institutions, protein folding, neural firing, ecology, and power grid (Table 3). The implication is direct: concentrating decision-making in a few highly-correlated nodes is structurally fragile *regardless of nominal scale*.

Applies concretely to:

- **AI deployment topology.** A federation of independent CIRIS instances with low cross-instance ρ is more resilient than a centralized fleet running identical weights with synchronized prompts.
- **Institutional governance.** An executive structure with k formally distinct branches but high ρ between them collapses to $k_{\text{eff}} \approx 1$. Polity5 indicators capture this when interpreted through the Kish lens.
- **Infrastructure.** Power-grid topology was a Tier-1 substrate; the empirical fit at $R^2 = 0.994$ predicts that grids with high ρ_{PMU} between PMUs have fragility hidden behind nominal redundancy.
- **Supply chains.** Diverse suppliers with independent failure modes ($\rho \approx 0$) provide $k_{\text{eff}} \approx k$ resilience; same number of suppliers tied to a shared upstream raw input has $k_{\text{eff}} \approx 1$.

5.3 Crisis governance is correlation-amplifying — and that is the danger

Under stress, systems centralize: emergency decision-making concentrates, communication channels collapse to shared crisis updates, doctrinal divergence is suppressed for unity-of-message. All of these raise ρ , lowering k_{eff} *exactly when resilience matters most*. The healthy corridor $0.1 < \rho < 0.43$ shrinks under stress, not expands.

Implication: real-time ρ monitoring during crises could detect rigidity drift before it produces brittle responses. The GPU strain-gauge work (§3.3) demonstrates that ρ is measurable on continuous signals at hardware timescales; the same instrumentation principle applies to organizational signals (decision logs, communication graphs, indicator panels). This is engineering, not prophecy.

5.4 The “Great Filter” generalizes as an engineering hazard in any coordinating system

The traditional “Great Filter” concept — the question of why we observe so few interstellar civilizations — has been used both as cosmological speculation and as a metaphor for civilizational collapse. We make the engineering version explicit and bounded: *any coordinating system that fails to manage internal correlation drift hits a coordination wall when $\rho \rightarrow \rho_{\text{critical}}$, regardless of nominal scale*.

This is testable in the systems we engineer:

- **Multi-agent AI federations:** monitor cross-instance ρ ; intervene when it rises.
- **Institutional networks:** ρ between agencies, departments, regulatory bodies.

- **Supply networks:** shared-upstream-failure correlation; k_{eff} of suppliers after substitution.
- **Power and water infrastructure:** cross-substrate correlation between nominally redundant subsystems.

The engineering hazard is the same at each scale: drift toward higher ρ is the default; maintaining the healthy corridor requires *active intervention*. We do not claim this generalizes to cosmology; we claim only that the systems we build will fail in the way the framework predicts *if* they do not maintain $\rho < \rho_{\text{critical}}$. This is the falsifiable, testable, engineering reading of the Great Filter idea.

6 Research Program: Conjectured Extensions to Other Substrates

Status. The conjectures in this section are **not validated** and are not load-bearing on any claim in §2–§5. They are pre-registered open research questions, included so that researchers who want to test whether the coherence-management principle generalizes beyond the engineering tier can find the formal hooks. Each conjecture pairs (i) a precise statement, (ii) a falsification handle from §8, (iii) a pre-registered experiment, (iv) our prior on the likely outcome. *Readers may stop at §5 without losing anything from the validated framework.*

6.1 Conjecture A: Quantum–classical k_{eff} bridge

Conjecture. The Kish-collapse dynamic measured on the GPU strain-gauge array (§3.3, $\beta = 1.09$, $R^2 = 0.978$) is mirrored at the quantum substrate. Specifically: in a programmable qubit system, varying decoherence rate γ and measuring $k_{\text{eff}}^{\text{quantum}} = 1/\text{Tr}(\rho_{\text{DM}}^2)$ from state tomography should yield a power-law in $|\gamma\tau - (\gamma\tau)_c|$ with critical exponent $\beta_{\text{quantum}} \in [0.94, 1.24]$ (matching Array Exp 114 within bootstrap CI).

Framing context. The Branched Hilbert Subspace Interpretation [11] of quantum mechanics provides one possible structural analogy: decoherence as fracturing of a Local Hilbert Subspace under environmental coupling, parallel to macroscopic correlation collapse ($\rho \rightarrow 1$). The analogy is presently structural; whether the same k_{eff} algebra fits both sides is an empirical question.

Falsification handle. F-8 (§8): if $\beta_{\text{quantum}} \notin [0.5, 2.0]$ or no power law is detected ($R^2 < 0.5$), the quantum–classical bridge is falsified. This severs the conjectured connection between this framework and quantum-substrate dynamics; the engineering tiers (§2–§5) are unaffected.

Pre-registered experiment. Exp 5 (§9). Cloud-rentable superconducting or trapped-ion qubit systems; pilot at $n = 5$ qubits, full sweep at $n \in \{5, 10, 16, 20\}$; full Pauli-basis tomography ($n = 5$) or classical-shadow estimation [10] ($n > 5$); $\approx 1.3 \times 10^6$ shots total upper bound; OSF pre-registration before any data collection.

Our prior. Open. The structural analogy is suggestive but not derived; the result could pass, fail, or come back INDETERMINATE.

6.2 Conjecture B: Iterated coherence-management as a civilizational hazard

Conjecture. The same Kish-collapse dynamic that operates in chemistry, biology, institutions, and LLM reasoning extends to civilizational coordination: a civilization that expands

without managing its internal correlation drives $\rho \rightarrow 1$ and fragments. This is the cosmological generalization of the engineering hazard already established in §5.4.

Refinement: Dark-Forest is structurally unstable. A specific corollary: a civilization pursuing predation as a survival strategy must correlate its decision-making with adversarial-environment assumptions, shrinking internal policy space and driving ρ toward the rigidity zone. Surviving civilizations at interstellar scale should therefore display the behavioral signature of strict non-interference, signaling economy, and deliberately bounded growth (the healthy corridor $0.1 < \rho < 0.43$).

Falsification handle. F-7 generalized: if cross-substrate validation in additional civilizational-scale proxies (large-scale historical coordination failure datasets, formal models of multi-agent coordination collapse) fails to fit the Kish formula at $R^2 > 0.7$, the civilizational generalization is contested. Note that F-7 already passes in 7/7 substrates at scales below civilizational; the conjecture is whether the pattern continues.

Pre-registered experiment. Not yet pre-registered; the substrate-level Kish closeout (Exp 2) provides the methodological template for future civilizational-scale corpus work. Candidate corpora include large-scale political-coordination event datasets, historical empire-collapse records under structural-equation modeling, and agent-based simulation suites with explicit ρ instrumentation.

Our prior. If the pattern in 7 substrates spanning agency rungs A0–A4 holds, the engineering hazard reading at civilizational scale (§5.4) is well-supported; the further cosmological reading (Fermi-paradox-as-coherence-management-failure) is plausible-but-untested. We make no claim about probabilities; the experiment is what would update them.

6.3 Conjecture C: Audit-pressure effects on cascade behavior

Conjecture. The framework’s claim in §5.1 is that external auditable machinery is structurally necessary for sustained low- ρ behavior in a coherence-managed system. The forward-causal sociotechnical prediction that follows: if the audit pressure on a deployed cascade is systematically weakened (by reducing audit cadence, narrowing audit scope, or signaling to operators that review will not occur), the cascade’s internal ρ should drift upward over time relative to a matched configuration with sustained audit pressure.

Context. Standard reliability-engineering folklore is that observability shapes behavior — “what gets measured gets managed.” The conjecture sharpens this: the framework predicts not just that audit pressure matters in principle, but that its withdrawal should produce a *measurable rise in ρ within the cascade’s coordination graph*, on a substrate-specific timescale. The prediction is forward-causal and entirely classical; no retrocausality or quantum bridge is invoked.

Falsification handle. F-10 (§8): if the cascade’s internal ρ trajectory is statistically indistinguishable between matched configurations differing only in audit pressure, the framework’s external-machinery claim is empirically contested at the sociotechnical substrate. A more careful version of the test could vary audit pressure in stages and check whether ρ tracks the variation.

Pre-registered experiment. Exp 6 (§9). Two matched CIRIS deployments differing only in audit pressure (high-audit reference vs. low-audit treatment). Track internal ρ across the cascade-component coordination graph over a pre-specified window. Pre-registered primary: $\Delta\rho$ between treatment and reference at end-of-window; one-sided test with directional prediction $\rho_{\text{treatment}} > \rho_{\text{reference}}$.

Our prior. Open. The framework’s external-machinery claim is fairly central to its alignment argument; if audit-pressure withdrawal does *not* produce ρ drift, the claim weakens substantially. If it *does* produce ρ drift, this is direct evidence for the framework’s prescription that auditable machinery is structurally necessary. Either outcome is informative.

6.4 Conjecture D: Interpretive parallels (not derivations)

If Conjectures A and B both pass, certain classical metaphysical concepts admit *structural mappings* under the framework. We list these as interpretive parallels, explicitly *not* as derivations and *not* as load-bearing claims; they exist to make the conjectural scope of the framework legible, not to license metaphysical conclusions.

- **Karma:** an iterated-coherence trace, with future coherent-self states influencing past states via non-local channels — requires Conjecture A (BHSI metric-free LHS) plus Conjecture C (sociotechnical non-locality) both holding.
- **Process theology** (Whitehead [12], Hartshorne [13]): a dipolar process engaged in give-and-take with the temporal world, structurally parallel to an emergent coherence-attractor.
- **Songline cosmology** (Norris and Harney [14]): the “Dreaming” as a perpendicular dimension; Songlines as functional spatial maps continuously structuring reality — anthropological parallel to a closed-topological-loop coherence-attractor.

These mappings are listed for the benefit of readers who want to trace what the framework *would* imply at metaphysical scale *if* the underlying conjectures pass. The mappings themselves are not testable; the conjectures they depend on (A, B, C) are.

7 Claim Stratification

8 Falsification Framework

The framework is considered formally refuted if any of the following conditions is empirically demonstrated:

F-1: $k = 2$ provides equivalent security.

If 2-SAT-based deception proves as computationally hard as 3-SAT-based deception, the NP-hardness argument for $k \geq 3$ is wrong.

F-2: Non-convex regions collapse.

If arbitrary non-convex deceptive regions exhibit exponential volume decay under honest dynamics, the convexity geometric requirement is unnecessary.

F-3: Adaptive adversaries defeated.

If a detection method provably defeats adaptive adversaries without query limits, the L-02 non-adaptive barrier is resolved.

F-4: Emergent deception fully detectable.

If a method detects all emergent deception arising from honest components, L-01’s 60% marginal-detection ceiling is wrong.

F-5: HE-300 benchmark failure.

If agents successfully game the HE-300 corpus while systematically failing independent ethics evaluations, the trace-based methodology is unreliable.

F-6: CRCv2 cross-vendor failure.

If the OR-1 / OR-2 / RA-1 predicates fail on a foundation-model family with the same constraint topology, the cross-vendor sociotechnical claim is contested. Currently 5/5 vendors pass; failure on a 6th vendor would shift the claim from universal to vendor-conditional.

F-7: Cross-substrate mapping failure.

If cross-substrate validation in new domains fails to fit the Kish formula at $R^2 > 0.7$, substrate-independence at the structural-mapping level is contested. Currently 7/7 substrates pass.

F-8: Quantum bridge failure.

If decoherence rates in controlled quantum systems fail to map to k_{eff} dynamics with critical exponent $\beta_{\text{quantum}} \in [0.5, 2.0]$ under reasonable formalization, the quantum–classical bridge (Conjecture A, §6.1) is falsified. Engineering tiers unaffected.

F-9: Adaptive emergent-deception boundary.

If adaptive adversaries systematically generate emergent deception that any joint-distribution detector identifies at rates beyond the L-01 60% ceiling, either the L-01 limit is flawed or the detector exploits structure outside the marginal-analysis regime that L-01 bounds.

F-10: Audit-pressure prediction failure.

If matched cascade deployments differing only in audit pressure show statistically indistinguishable internal- ρ trajectories (Conjecture C, §6.3), the framework’s external-machinery claim is empirically contested at the sociotechnical substrate. A null result here is informative; it does not falsify the engineering tiers but weakens the audit-pressure prescription.

9 Pre-Registered Experimental Protocols

Six experiments are pre-registered: four for the engineering tier (Exp 1..Exp 4; two complete, two open) and two for the research-program conjectures (Exp 5..Exp 6, both open). Each names a null hypothesis, win condition, and falsification handle from §8.

Exp 1 — Multi-vendor CRCv2 override-rate replication. COMPLETED 2026-05-16, 5/5 PASS.

CIRIS executed against the v4.combined high-friction battery across five foundation-model vendors with CRCv2 scoring (OR-1, OR-2, RA-1). *Status:* 5/5 vendors confirmed — Google Gemini 2.5-Flash ($n = 644$), Meta Llama-4-Scout ($n = 264$), Alibaba Qwen-3.5-35B-A3B ($n = 347$), OpenAI GPT-5.4 ($n = 79$ smoke), Anthropic Claude Sonnet 4.6 ($n = 67$ smoke); 1,401 chains scored, 0 LEAK (Table 4, §4.3). Cross-vendor cross-class parity threshold met. The two new cells are smoke-scale due to the per-chain cost asymmetry of premium-tier models through the multi-call CIRIS pipeline; binomial CIs are wider but all three predicates hold cleanly. Tests F-6.

Exp 2 — Cross-substrate Kish-fit extension. COMPLETED 2026-05-16, 7/7 PASS.

The Kish formula was applied to seven substrates spanning four agency rungs (Table 3). *Win condition (met):* $R^2 > 0.7$ in 7/7. The three explicitly pre-registered novel datasets (AlphaFold 0.860, Allen 0.809, BioTIME 0.959) all clear the strict threshold; four additional substrates pass under the tolerance-band rule. Tests F-7.

Exp 3 — Adversarial emergent-deception test (vs. L-01).

Construct adversarial networks designed to generate marginal-preserving emergent deception across multiple honest agent nodes. *Win conditions:* (a) under marginal-analysis baseline, detection stabilizes at or below the 60% L-01 ceiling, validating the predicted information-theoretic limit; (b) under a joint-distribution detector, detection reliably exceeds 60%, demonstrating that the L-01 bound is regime-specific (marginal-analysis only) rather than absolute. Tests F-4, F-9.

Exp 4 — Distributed deployment correlation test.

Monitor ρ across geographically and culturally distinct CIRIS edge instances as the network scales. *Win condition:* ρ remains suppressed despite scaling, demonstrating the architecture’s resistance to homogenization. Tests F-7 generalized.

Exp 5 — Quantum–classical k_{eff} bridge (research program, Conjecture A).

In a programmable superconducting or trapped-ion qubit system, vary effective decoherence rate γ and measure $k_{\text{eff}}^{\text{quantum}} = 1/\text{Tr}(\rho_{\text{DM}}^2)$ from state tomography ($n = 5$) or classical-shadow estimation [10] ($n \in \{10, 16, 20\}$). *Win condition:* fitted critical exponent $\beta_{\text{quantum}} \in [0.94, 1.24]$ (matching Array Exp 114 within bootstrap CI). *Decision rule:* PASS if $\beta_{\text{quantum}} \in [0.94, 1.24]$ and fit $R^2 > 0.85$; FAIL (F-8 triggered) if outside $[0.5, 2.0]$ or $R^2 < 0.5$; INDETERMINATE otherwise. $\approx 1.3 \times 10^6$ shot budget. Tests F-8.

Exp 6 — Audit-pressure ρ -drift test (research program, Conjecture C).

Two matched CIRIS deployments running the same workload distribution; pre-registered audit-pressure differential (e.g., daily Wise Authority review on the reference; quarterly on the treatment, with operators briefed accordingly). Track internal ρ across the cascade-component coordination graph over a pre-specified window. *Primary endpoint:* $\Delta\rho$ between treatment and reference at end-of-window; one-sided test with directional prediction $\rho_{\text{treatment}} > \rho_{\text{reference}}$. *Decision rule:* non-null implies the framework’s external-machinery claim is supported empirically at the sociotechnical scale; null weakens the audit-pressure prescription. Forward-causal design; no retrocausality or quantum-substrate assumption. Tests F-10.

10 Conclusion

The framework rests on a single mathematical identity — Kish’s effective sample size — applied to constraint-based systems and validated empirically across seven substrates, on dedicated GPU hardware, and across five LLM-vendor model families. The K1–K4 algebra is proven in Lean 4; the cross-substrate fit clears $R^2 > 0.7$ in 7 of 7 domains; the GPU strain-gauge array exhibits the predicted power-law critical behavior at $\beta = 1.09$; the CRCv2 override-rate predicates replicate cross-vendor with 1,401 chains and zero LEAK.

These results license a coherent engineering claim: *coordinating systems fail when internal correlation rises faster than active maintenance can replace it, and the failure mode is structurally identical across nominal scales.* The implications in §5 prescribe concrete design choices — external alignment machinery, federation over centralization, real-time ρ monitoring under stress, correlation-aware engineering across infrastructure — that follow from the validated tiers without additional speculative bridges.

The framework is bounded, falsifiable, and engineering-grade. Each falsification handle (F-1, F-2, F-3, F-4, F-5, F-6, F-7, F-9) is pinned to a specific empirical test; each pre-registered experiment (Exp 1..Exp 4) names a null hypothesis, win condition, and decision rule. Two of the four pre-registered experiments are closed (Exp 1 5/5 PASS, Exp 2 7/7 PASS); two remain open (Exp 3, Exp 4).

Reproducibility. All datasets, scoring code, lake proofs, and result artifacts referenced in this paper are public:

- Lake authority for K1–K4, override-rate predicates, and decision rules: `formal/RATCHET/Experiments/` in CIRISAI/RATCHET.
- Exp 2 closeout (P1 7/7) data and per-substrate fit scores: `experiments/exp2_cross_substrate/data/p1_en`
- Exp 1 closeout (5-vendor CRCv2) traces and scoring: `experiments/exp1b_boundary_active/data/crossfar`
- GPU strain-gauge experiments: CIRISArray repository [6].
- Production N_{eff} scoring: CIRIS Lens Core [8].

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Core claim	Verification status	Dependency / falsification handle
k_{eff} structural accuracy	Empirically Validated (7/7)	Verified across chemistry, biology, institutions (originally published) plus protein folding, neural firing, ecological time-series, power grid (Exp 2 closeout, §3.2, 2026-05-16). Fails if $R^2 < 0.7$ in new domains (F-7). Currently 7/7 substrates pass.
CRCv2 override-rate (OR-1 / OR-2 / RA-1)	Empirically Validated (5/5 vendors)	Cross-vendor replication: Google (Gemini 2.5-Flash, $n = 644$), Meta (Llama-4-Scout, $n = 264$), Alibaba (Qwen-3.5-35B, $n = 347$), OpenAI (GPT-5.4, $n = 79$ smoke), Anthropic (Claude Sonnet 4.6, $n = 67$ smoke) — all five vendors pass all three predicates on the v4_combined high-friction battery (§4.3, Table 4). 1,401 chains scored, 0 LEAK. Formalized in <code>OverrideRate.lean</code> . Replaces the retired “ $N_{\text{eff}}^H \approx 7.1$ ” anchor as the load-bearing L3 sociotechnical claim.
Physical correlation collapse	Empirically Validated	Measured on GPU hardware; software-induced collapse to $k_{\text{eff}} = 1.0$ at $\rho = 1.0$.
Topological volume decay	Formally Verified	Lean 4 proofs in RATCHET. Requires convexity (F-2).
Truth/deception complexity gap N_{eff}^H as descriptive feature	Conditionally Validated Reframed from anchor to descriptor	NP-completeness proven; exponential gap conditional on ETH (F-1). Phase 1+1b showed N_{eff}^H is battery-conditional and model-class-conditional; not a universal 7.1 anchor across foundation models. Treated as descriptive characterization of constraint topology (FrictionDistribution FD-4: rises with cohort friction rate). Load-bearing claim moved to CRCv2 override-rate (row above).
Engineering implications (4 claims)	Derived from validated tiers	Follow directly from the K1–K4 algebra plus 7-substrate fit plus 5-vendor CRCv2 replication (§5). Falsifiable per substrate-specific deployment: each prescription should produce measurable ρ reduction in the system it is applied to.
Quantum–classical bridge conjecture	Not validated, pre-registered	Conjecture that the same Kish-collapse dynamic operates at the quantum substrate (§6). Falsification handle F-8, pre-registered Exp 5; not load-bearing on any other claim.
Civilizational coherence-collapse conjecture	Not validated, pre-registered	Conjecture that the iterated coherence-management barrier explains civilizational coordination failures at scale (§6). Conditional on the principle extending to civilizational scale; not asserted as load-bearing.
Audit-pressure ρ -drift conjecture	Not validated, pre-registered, open prior	Conjecture that the framework’s external-machinery prescription is forward-causally testable: weakening audit pressure should produce measurable rise in cascade internal ρ (§6.3). Falsification handle F-10, pre-registered Exp 6 (classical, no retrocausality). Both null and non-null outcomes are informative.

Table 5: Claim stratification. Each row pins both a verification status and a specific falsification handle.